Project Proposal

A Refund Support Assistant Using Retrieval-Augmented Generation and QLoRA Fine-Tuning

Item	Detail
Stage	Stage 1: Business Objective, Data Understanding, and Proposed Methodology
Scope of this document	Task 1.1 (Define Project Scope) and Task 1.4 (Design Methodology)
Companion deliverable	EDA Notebook (Tasks 1.2 and 1.3)
Version / date	1.0 / 6 October 2026

1. Executive Summary

This proposal plans a domain-specific customer-support assistant for a single category, **REFUND**, of the Bitext customer-support dataset. The assistant receives a customer query as JSON and returns a structured JSON answer containing the detected intent, a response grounded in Standard Operating Procedure (SOP) documents, the SOP section used, and an escalation flag.

Three systems will be compared on the same held-out test set: a zero-shot **Baseline** model, **V1 Naive RAG** (baseline plus SOP retrieval), and **V2 Hybrid RAG** (the same retrieval plus a QLoRA-fine-tuned model that also routes intent). The retriever is frozen between V1 and V2, so V1 minus Baseline isolates the contribution of retrieval and V2 minus V1 isolates the contribution of fine-tuning.

Success is measured on format adherence, intent accuracy, ROUGE/BLEU against gold responses, hallucination frequency and consistency, with provisional numeric targets in Section 4.

2. Business Objective

Refund enquiries are frequent, repetitive and policy-bound, which makes them a good fit for automation. Off-the-shelf language models handle them poorly: they refuse or deflect, invent policy details, and return free text that downstream systems cannot parse. The business objective is a small, cheap-to-run model that answers refund questions accurately from company SOPs and returns machine-readable output.

- **Accuracy:** identify the customer's refund intent correctly and answer from SOP content only.
- **Reliability:** always return valid, schema-conforming JSON so the output can drive routing and ticketing.
- **Safety:** reduce unsupported (hallucinated) policy claims and escalate when the SOP does not cover the request.
- **Efficiency:** train and serve on a single commodity GPU using parameter-efficient fine-tuning.

3. Target Use Case and Project Boundaries (Task 1.1.1)

3.1 Selected support category

Category: REFUND. One category keeps the SOP corpus small enough to review for conflicts, keeps class balance tractable, and lets the fine-tuning effect be measured cleanly. In-scope intents:

Intent	Customer need	Governing SOP section	Typical escalation
`get_refund`	Request a refund for an order	SOP-REF-02: Requesting a refund	When eligibility is unclear
`check_refund_policy`	Ask about refund rules, windows and conditions	SOP-REF-01: Refund policy and eligibility	Rarely
`track_refund`	Check status or timing of an existing refund	SOP-REF-03: Refund status tracking	When refund is overdue per SOP
`out_of_scope`	Any query outside refunds (fallback label added by this project)	None (`sop_reference` is null)	Redirect or hand off

SOP identifiers are working names for the SOP documents supplied with the project; they will be aligned to the real document structure during EDA.

3.2 Input and output contract

Input (one JSON object per request):

```
{
  "request_id": "r-0001",
  "query": "I am waiting for a refund of {{Currency Symbol}}{{Refund Amount}} for order {{Order Number}}"
}
```

Output (exactly one JSON object, no text outside it; illustrative values):

```
{
  "intent": "track_refund",
  "category": "REFUND",
  "response": "I understand you are waiting for your refund. You can follow its status ...",
  "sop_reference": "SOP-REF-03",
  "escalate_to_human": false
}
```

Field	Type	Constraint
`request_id`	string	Optional on input; echoed by the service wrapper, not generated by the model
`query`	string	Required on input, 1 to 500 characters, English, single turn
`intent`	string (enum)	One of `get_refund`, `check_refund_policy`, `track_refund`, `out_of_scope`
`category`	string (enum)	`REFUND` or `OUT_OF_SCOPE`, consistent with `intent`
`response`	string	1 to 1,200 characters; every factual claim must be supported by the SOP; Bitext template slots such as {{Order Number}} are kept as-is, never filled with invented values
`sop_reference`	string or null	Pattern `SOP-REF-NN`, or null when `intent` is `out_of_scope`
`escalate_to_human`	boolean	True when the SOP requires human handling or no SOP section applies

Output validity is checked with a JSON Schema (draft 2020-12) using `additionalProperties: false`, `required` for all five output fields, and enums for `intent` and `category`. The same schema is used for training-data validation and for the Format Adherence metric.

3.3 Boundaries

In scope	Out of scope
Single-turn English text queries about refunds	Multi-turn dialogue, voice, or non-English input
Three refund intents plus an `out_of_scope` fallback	Other Bitext categories (orders, accounts, payments, delivery and so on), which are routed to `out_of_scope`
Answers grounded in the project SOP documents	Executing refunds, looking up real orders, or integrating with live systems
Dataset of 1,000 to 5,000 examples after cleaning	Handling real personal or payment data; legal or financial advice
Training on a single 16 GB GPU with QLoRA	Full-parameter fine-tuning, production deployment, or load testing

4. Success Criteria (Task 1.1.2)

Targets apply to **V2 Hybrid RAG** on the held-out test set unless stated. They are provisional: they will be re-calibrated once the zero-shot Baseline is measured in the next stage, and any change will be documented with the reason.

Objective	Metric and definition	Target
Format Adherence	Share of outputs that parse as JSON, pass schema validation, and contain no text outside the object	≥ 98%, and not lower than V1
Intent Accuracy	Exact match of predicted `intent` against gold; macro-F1 across the four labels	Accuracy ≥ 90%; macro-F1 ≥ 0.88
Text overlap (ROUGE / BLEU)	ROUGE-1, ROUGE-2, ROUGE-L F1 and smoothed BLEU of `response` against the **gold Bitext response**	ROUGE-L ≥ 15% relative gain over V1; BLEU no lower than V1; paired bootstrap 95% CI excludes zero
Hallucination Reduction	Share of responses containing at least one claim not supported by the SOP	≤ 5% absolute and ≥ 50% relative reduction versus the zero-shot Baseline
Consistency	Share of queries with identical `intent` and `escalate_to_human` across 3 runs and paraphrases	≥ 95%
Retrieval quality	Recall@3 and MRR of the correct SOP section on the validation set	Recall@3 ≥ 0.90; MRR ≥ 0.80
Feasibility	Peak GPU memory during QLoRA training; training completes in a single session	≤ 16 GB; no out-of-memory failures

Project success means V2 meets at least five of the seven targets and shows statistically significant improvement over V1 on ROUGE-L and hallucination frequency. A negative or null result on any component is still reported, with analysis, rather than hidden.

Attribution design. Retrieval contribution = V1 minus Baseline. Fine-tuning contribution = V2 minus V1. Total improvement = V2 minus Baseline, which is the sum of the two absolute differences. All three are reported with confidence intervals.

5. Data Understanding and Preparation Plan

The detailed analysis is in the EDA notebook (Tasks 1.2 and 1.3). This section records the plan that the methodology depends on.

- **Sources:** the Bitext customer-support dataset (instruction, category, intent, response) filtered to the REFUND category, and the project SOP documents.

- **Quality checks:** completeness, formatting inconsistencies, exact duplicates, and near-duplicates found by normalised deduplication (lower-casing, whitespace and placeholder normalisation); verification that each response matches its labelled intent.
- **Sizing:** profile prompt and response token lengths, then downsample to the 1,000 to 5,000 range while keeping the three intents balanced. About 10% of the final set will be `out_of_scope` examples drawn from other categories.
- **Splits:** stratified 70/15/15 train, validation and test, created before any retrieval or fine-tuning, with near-duplicate groups kept inside a single split to prevent leakage.
- **SOP review:** TF-IDF cosine similarity between SOP sections to flag overlapping or conflicting rules before they are indexed.

6. Proposed Methodology (Task 1.4)

6.1 Experimental design

System	Retrieval	Model	Question it answers
Baseline	None	Base instruct model, zero-shot	What does the model do on its own?
V1 Naive RAG	Dense top-3 over SOP chunks	Base instruct model	What does retrieval add?
V2 Hybrid RAG	Same retriever as V1 (frozen)	Base model + QLoRA adapter	What does fine-tuning add on top of retrieval?

All systems use the same test queries, prompt wrapper, decoding settings and metric code. Holding the retriever fixed between V1 and V2 is deliberate: otherwise a retrieval change would be confounded with the fine-tuning effect.

6.2 Baseline model selection (Subtask 1.4.1)

Selected model: Qwen2.5-1.5B-Instruct. The justification rests on three factors.

- **Hardware constraints.** The target is one 16 GB GPU (for example a free-tier Colab T4). At 4-bit precision a 1.5B-parameter model needs roughly 1 to 1.5 GB for weights, leaving ample room for LoRA adapters, optimiser state, activations and a 1,024-token context at batch size 4. Larger 3B to 4B models would fit but leave little headroom and slow iteration.
- **Dataset characteristics.** REFUND queries are short and responses are moderate in length. A prompt of query plus three SOP chunks plus a JSON answer is expected to fit within about 1,000 tokens (to be confirmed by the token profiling in EDA), well inside the model's context window.
- **Fine-tuning feasibility.** The model family natively uses the ChatML template, so training data matches the pretrained chat format. It is released under Apache 2.0, requires no access approval, and is supported by the `transformers`, `peft`, `trl` and `bitsandbytes` stack.

Candidate	Size	Chat template	Fit for 16 GB QLoRA	Decision
Qwen2.5-1.5B-Instruct	1.5B	ChatML	Comfortable	**Selected**
Qwen2.5-0.5B-Instruct	0.5B	ChatML	Very comfortable	Fallback if memory or time is tight
Llama-3.2-1B-Instruct	1B	Llama 3 format	Comfortable	Not selected: gated access, non-ChatML template
Phi-3-mini-4k-instruct	3.8B	Phi-3 format	Tight	Not selected: slower iteration, little headroom
TinyLlama-1.1B-Chat	1.1B	Zephyr-style	Comfortable	Not selected: weaker instruction following

Model properties are stated from the published model cards as understood at the time of writing and will be re-verified when the notebooks are built.

6.3 Retrieval strategy (Subtask 1.4.2)

1. **SOP preparation.** Split SOP documents on section headings, then into chunks of about 200 to 300 tokens with roughly 40 tokens of overlap. Each chunk carries metadata: `sop_id`, section title and mapped intent.
2. **Embedding model.** `BAAI/bge-small-en-v1.5` (384-dimension vectors, 512-token input limit). The limit comfortably exceeds the chunk size; a 256-token model would silently truncate chunks.
3. **Vector storage.** FAISS `IndexFlatIP` over L2-normalised vectors, giving exact cosine-similarity search. The corpus is only a few hundred chunks, so exact search is fast and an approximate index is unnecessary. The index and chunk metadata are persisted to disk and versioned.
4. **Retrieval workflow.** (1) Embed the query. (2) Retrieve the top-3 chunks by cosine similarity. (3) If the best score is below a threshold tuned on the validation set, flag the query as likely out of scope. (4) Pack the chunks into the prompt with their `sop_id` labels. (5) Generate the JSON answer.
5. **Retrieval ablation.** Before any generation experiment, compare dense-only retrieval with dense plus BM25 fused by Reciprocal Rank Fusion. Select one retriever, then **freeze it** for both V1 and V2.
6. **Retrieval evaluation.** On the validation set, each query's relevant chunks are those mapped to its gold intent. Report Recall@1, Recall@3, Recall@5, MRR and context precision, with a target of Recall@3 ≥ 0.90.

6.4 Fine-tuning strategy (Subtask 1.4.3)

Method: QLoRA. The base model is loaded in 4-bit NF4 quantisation with double quantisation and frozen. Low-rank adapters are trained on top, with fp16 compute on a T4 (bf16 where the GPU supports it).

Setting	Value
LoRA rank / alpha / dropout	r = 16, alpha = 32, dropout = 0.05 (small sweep over r in {8, 16, 32} if time allows)
Target modules	`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj` (all attention and MLP projections)
Optimiser and schedule	`paged_adamw_8bit`, learning rate 2e-4, cosine decay, 3% warm-up, weight decay 0.01, gradient clipping 0.3
Batching	Per-device batch 4 with gradient accumulation 4 (effective batch 16); gradient checkpointing on
Sequence length / epochs	Maximum 1,024 tokens; up to 3 epochs with early stopping (patience 3 evaluations)
Loss	Completion-only: loss is computed on the assistant JSON tokens, not on the system or user prompt
Reproducibility	Fixed seed (42); library versions, config and data hashes saved with each run

ChatML structure. Each training example is a three-turn conversation. The user turn contains the retrieved SOP context and the query, and the assistant turn is the gold JSON object, so the model learns to ground its answer in SOP text and to emit valid JSON.

```
<|im_start|>system
You are a refund-support assistant. Answer only from the SOP context. Reply with one JSON
object that follows the schema. If the request is not about refunds, set intent to out_of_scope.<|im_end|>
<|im_start|>user
SOP CONTEXT:
[SOP-REF-03 section 2] ...retrieved chunk text...
QUERY: I am waiting for a refund of {{Currency Symbol}}{{Refund Amount}}<|im_end|>
<|im_start|>assistant
{"intent": "track_refund", "category": "REFUND", "response": "...", "sop_reference": "SOP-REF-03",
"escalate_to_human": false}<|im_end|>
```

The gold JSON is assembled from Bitext fields: `intent`, `category` and `response` come from the dataset, `sop_reference` from a deterministic intent-to-SOP mapping, and `escalate_to_human` from SOP rules. Out-of-scope examples use `intent = out_of_scope` and a polite redirect response.

Checkpointing. Evaluate and save every 50 steps, keep the three most recent checkpoints plus the best by validation loss, write checkpoints to persistent storage (such as Google Drive) so a session disconnect does not lose progress, and support `resume_from_checkpoint`. The final deliverable is the best adapter, saved separately from the base model.

Logging. Log to TensorBoard (or Weights & Biases): training and validation loss, learning rate, gradient norm, tokens per second and peak GPU memory. At each evaluation, generate responses for a fixed set of 10 validation queries and store them, so qualitative progress and JSON validity can be inspected over time.

6.5 Evaluation framework (Subtask 1.4.4)

Protocol. Evaluate all three systems on the same held-out test set with greedy decoding (temperature 0), a 256-token generation budget and the ChatML stop token. A budget that is too short truncates JSON and would unfairly depress every metric, so it will be checked for truncation before results are accepted.

Metric	Procedure
Format Adherence	Parse each output with `json.loads`; validate against the JSON Schema; reject any output with text outside the object. Report the pass rate and a breakdown of failure types (invalid JSON, missing field, bad enum, extra text).
Intent Accuracy	Compare predicted `intent` with the gold label by exact match. Report accuracy, per-class precision, recall and F1, macro-F1, and a confusion matrix. Unparseable outputs count as incorrect.
Consistency	Run each test query three times with sampling (temperature 0.7) and also score 3 paraphrases per query for a 100-query subset. A query is consistent if `intent` and `escalate_to_human` are identical across runs. Also report mean pairwise ROUGE-L between `response` texts.
Hallucination Frequency	Claim-level check against the SOP. Step 1: automatically extract numbers, durations and policy statements from each response and check that they appear in the SOP. Step 2: judge the remaining claims with a rubric-based LLM judge. Step 3: manually audit a random sample of 50 responses to calibrate the judge and report agreement. A response is hallucinated if it has at least one unsupported claim.
ROUGE / BLEU	Score `response` against the **gold Bitext response**, never against the intent label. Normalise template slots (for example {{Order Number}}) before scoring. Report ROUGE-1/2/L F1 and sacreBLEU with smoothing, with precision and recall shown separately because answer length affects F1.
Statistical testing	Paired bootstrap (5,000 resamples) for 95% confidence intervals and Wilcoxon signed-rank tests for each system pair, since all systems share the same queries.

The report will present Baseline, V1 and V2 side by side, the attribution of improvement to retrieval versus fine-tuning, an error analysis of the worst-performing intents, and example outputs for each

system.

7. Risks and Mitigations

Risk	Mitigation
Metric validity: ROUGE computed against the wrong reference (for example intent labels) gives misleading rankings	Always score against gold responses; add a unit test that checks the reference field before scoring; pair ROUGE/BLEU with intent, format and hallucination metrics
Bitext responses are templated, so overlap metrics have a low ceiling	Treat ROUGE/BLEU as one signal among five; judge correctness through intent accuracy and SOP grounding
Small dataset and near-duplicates cause overfitting or train-test leakage	Normalised deduplication before splitting; keep near-duplicate groups in one split; early stopping on validation loss
SOP documents are thin or contain conflicting rules	TF-IDF similarity review; manual resolution; version the SOP corpus
GPU memory limits or session disconnects	4-bit quantisation, gradient checkpointing, small batch with accumulation; persistent checkpoints; fall back to Qwen2.5-0.5B-Instruct
Truncated or unstructured outputs	Adequate token budget and stop token; fine-tuning on JSON targets; optional JSON-constrained decoding as an ablation
Hallucination judge is itself biased	Combine automatic SOP checks, a rubric-based judge and a 50-sample manual audit

8. Deliverables and Next Stages

Stage	Deliverable	Exit criterion
1 (this stage)	Project Proposal PDF; EDA Notebook	Scope, schema, targets and methodology agreed; clean, split dataset ready
2	Baseline and V1 Naive RAG pipelines; retrieval evaluation	Baseline metrics measured; retriever selected and frozen; targets re-calibrated
3	QLoRA fine-tuning notebook; trained adapter (V2)	Training completes with logged, resumable checkpoints; best adapter saved
4	Comparative evaluation and final report	All three systems scored; attribution analysis and recommendations documented

9. Assumptions and Open Items

- REFUND is the selected category; the same design applies to another category with a different SOP set.
- A single 16 GB GPU is available. If more memory is available, the 3B model becomes a viable alternative and would be added as a documented ablation.
- SOP documents are supplied with the project; their real structure may change the SOP identifiers and chunking details.
- Numeric targets are provisional and will be revisited after the Baseline is measured.